LISHAN YANG

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RESEARCH INTERESTS

System Reliability, HPC, Autonomous Driving Systems, GPUs, Measurement

ACADEMIC APPOINTMENTS

Assistant Professor Department of Computer Science, George Mason University Since June 2022

EDUCATION

Ph.D. in Computer Science	William & Mary	June 2022
B.E. in Computer Science	University of Science and Technology of China (USTC)	June 2016

MEMBERSHIPS

- IFIP Working Group 10.4 (Dependable Computing and Fault Tolerance), elected in 2025
- IFIP Working Group 7.3 (Computer System Modeling), elected in 2026
- ACM Lifetime Member
- IEEE Member

AWARDS

2026 **Best Reviewer Award**, ACM SIGMETRICS

2026 **Distinguished Reviewer**, ACM SIGPLAN Symposium on Principles and Practice of Parallel Programming (PPoPP)

2025 **Outstanding Teaching Award** (Tenure-Line), Department of Computer Science, George Mason University

2024 **Best Paper Finalist**, The IEEE International Conference on Cluster Computing (CLUSTER)

2024 **Best Reviewer Award**, The International Conference on Performance Engineering (ICPE)

2022 **SPEC Kaivalya Dixit Distinguished Dissertation Award**

2022 **Graduate Park Award**, Department of Computer Science, William & Mary

INVITED TALKS

2026 The Sixth Workshop on Benchmarking and Performance of Machine Learning Workloads on Emerging Hardware (MLBENCH)
Characterizing and Enhancing the Resilience of Generative Large Language Models against Soft Errors

2024 Institute of Science Tokyo
Reliable Operation of Heterogeneous Systems: From Foundation to Practical Fortification

2024 The Virginia Polytechnic Institute and State University (Virginia Tech)
Reliable Operation of Heterogeneous Systems: From Foundation to Practical Fortification

TUTORIALS

My students are underscored.

2026 The Annual IEEE/IFIP International Conference on Dependable Systems and Networks (DSN)
Fault Injection for ML Applications: From DNNs to LLMs
Karthik Pattabiraman, Bo Fang, **Lishan Yang**, Abraham Chan, Yu Sun

2026 George Mason University & University of Florida
Vulnerability Discovery of Autonomous Driving Systems
Guanpeng Li, **Lishan Yang**, Bohan Zhang, Daehyun Lee

2026 Chinese American Educational Research & Development Association International Conference (CAERDA)
Machine Learning for Quantitative Educational Research: Data Analytical Approaches
Chuang Wang, Scarlet Zhang, Jinfa Cai, Ningfang Mi, Chiu C. Tan, Yu Wang, **Lishan Yang**

2025 Chinese American Educational Research & Development Association International Conference (CAERDA)
AI for Educational Research
Chuang Wang, **Lishan Yang**, Ningfang Mi, Scarlett Zhang, Kyle Cox, Yu Wang, Chiu Tan, Qiao Liu

RESEARCH FUNDING

2026 Amazon Research Award (Build on Trainium, Spring 2026)
STRATA: Adaptive Roll-Forward Recovery with Hierarchical Checkpoint/Restart for Trainium Post-Training
Role: Co-PI. Total: $200,000 AWS Credits
With Dr. Bo Fang (lead-PI, University of Texas, Arlington)

2024-2027 NSF (National Science Foundation) #CCF-2402940
Collaborative Research: SHF: Medium: End-to-End Resilience in Autonomous Driving Systems: Strategic Vulnerability Assessment and Mitigation
Role: lead-PI. Total: $927,747, my share: $338,546
With Dr. Homa Alemzadeh (University of Virginia) and Dr. Evgenia Smirni (William & Mary)

2024-2028 NSF (National Science Foundation) #OAC-2417718
Collaborative Research: CyberTraining: Implementation: Medium: AI4EDU: Cloud Infrastructure-Enabled Training for AI in Educational Research and Assessment
Role: PI. Total: $999,969, my share: $184,360
With Dr. Ningfang Mi (lead-PI, Northeastern University), Yu Wang (Temple University), Chiu C. Tan (Temple University), Chuang Wang (University of North Carolina at Charlotte)

2024-2027 NSF (National Science Foundation) #OAC-2410856
Collaborative Research: Elements: MELIOREM: An Integrated Evaluation Cyberinfrastructure towards Safe and Dependable Autonomous Driving Systems
Role: PI. Total: $589,751, my share: $279,674
With Dr. Guanpeng Li (lead-PI, University of Iowa)

2024-2025 Commonwealth Cyber Initiative (CCI), #HC-3Q24-047, Research in Supply Chain Cybersecurity
Securing the Machine Learning Components of Autonomous Systems: Risk Assessment and Mitigation
Role: co-PI. Total: $100K, my share: $50K
With Dr. Evgenia Smirni (lead-PI, William & Mary)

PUBLICATIONS

My students are underscored.

[1] Daehyun Lee, Yu Sun, Yiyang Lu, Anna Schmedding, Homa Alemzadeh, Evgenia Smirni, **Lishan Yang**, "DiscoFI: Discovering Safety-Critical Faults of Autonomous Driving Systems through Two-Phase Fault Injection", ACM SIGMETRICS 2027, to appear. (Acceptance rate: 17.7%)

[2] Michael Mogilevsky, Hazem Zaky, Yu Sun, Mingkai Zheng, **Lishan Yang**, Zhao Zhang, "Quantifying Performance Variability in GPU Clusters", IEEE Transactions on Parallel and Distributed Systems (TPDS), 2026.

[3] Yu Sun, Zachary Coalson, Shiyang Chen, Hang Liu, Zhao Zhang, Sanghyun Hong, Bo Fang, **Lishan Yang**, "Demystifying the Resilience of Large Language Model Inference: An End-to-End Perspective", In Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis (SC) 2025: pp. 1127-1144. (Acceptance rate: 21.2%)

[4] Yu Sun, Zhu Zhu, Cherish Mulpuru, Roberto Gioiosa, Zhao Zhang, Bo Fang, **Lishan Yang**, "FT2: First-Token-Inspired Online Fault Tolerance on Critical Layers for Generative Large Language Mod-

els", The 34th ACM International Symposium on High-Performance Parallel and Distributed Computing (HPDC) 2025: 7:1-7:14. (Acceptance rate: 18.8%)

[5] Cheng Chen, Grant Xiao, Daehyun Lee, **Lishan Yang**, Evgenia Smirni, Homa Alemzadeh, Xugui Zhou, "Safety Interventions against Adversarial Patches in an Open-Source Driver Assistance System", the 55th Annual IEEE/IFIP International Conference on Dependable Systems and Networks (DSN) 2025: pp. 599-608. (Acceptance rate: 20.2%)

[6] **Lishan Yang**, George Papadimitriou, Dimitrios Sartzetakis, Adwait Jog, Evgenia Smirni, Dimitris Gizopoulos, "GPU Reliability Assessment: Insights Across the Abstraction Layers", The IEEE International Conference on Cluster Computing (CLUSTER) 2024: pp. 1-13 (Acceptance rate: 28.2%, Best Paper Finalist)

[7] Anna Schmedding, **Lishan Yang**, Adwait Jog and Evgenia Smirni, "Aspis: Lightweight Neural Network Protection Against Soft Errors", The 35th IEEE International Symposium on Software Reliability Engineering (ISSRE) 2024: pp. 248-259 (Acceptance Rate: 25.9%)

[8] Anna Schmedding, Philip Schowitz, Xugui Zhou, Yiyang Lu, **Lishan Yang**, Homa Alemzadeh and Evgenia Smirni, "Strategic Resilience Evaluation of Autonomous Vehicle Neural Networks Against Transient Faults", 43rd International Conference on Computer Safety, Reliability and Security (SafeComp) 2024: pp. 33-48 (Acceptance Rate: 23.5%)

[9] Bohan Zhang, **Lishan Yang**, Hui Xu, Guanpeng Li, "Investigating the Impact of High-Level Software Design on Low-Level Hardware Fault Resilience", In 2023 53rd Annual IEEE/IFIP International Conference on Dependable Systems and Networks-Supplemental Volume (DSN 2023): pp. 163-167 (Disrupt 23 track)

[10] Anna Schmedding, Riccardo Pinciroli, **Lishan Yang**, Evgenia Smirni, "Epidemic Spread Modeling for COVID-19 Using Cross-fertilization of Mobility Data", IEEE Transactions on Big Data (TBD), 9(5), pp.1260-1275, 2023.

[11] Anna Schmedding, **Lishan Yang**, Riccardo Pinciroli, Evgenia Smirni, "GeoSpread: an Epidemic Spread Modeling Tool for COVID-19 Using Mobility Data", ACM International Conference on Information Technology for Social Good (GoodIT) 2022: pp. 125-131.

[12] Xugui Zhou, Anna Schmedding, Haotian Ren, **Lishan Yang**, Philip Schowitz, Evgenia Smirni, Homa Alemzadeh, "Strategic Safety-Critical Attacks Against an Advanced Driver Assistance System", 52th IEEE/IFIP International Conference on Dependable Systems and Networks (DSN) 2022: pp. 79-87.

[13] Riccardo Pinciroli, **Lishan Yang**, Jacob Alter, Evgenia Smirni, "Lifespan and Failures of SSDs and HDDs: Similarities, Differences, and Prediction Models", IEEE Transactions on Dependable and Secure Computing, 20(1), pp. 256-272, 2022.

[14] **Lishan Yang**, Bin Nie, Adwait Jog, Evgenia Smirni, "Enabling Software Resilience in GPGPU Applications via Partial Thread Protection", IEEE/ACM 43rd International Conference on Software Engineering (ICSE) 2021: pp. 1248-1259. (Acceptance Rate: 22%)

[15] **Lishan Yang**, Bin Nie, Adwait Jog, Evgenia Smirni, "SUGAR: Speeding Up GPGPU Application Resilience Estimation with Input Sizing", Proceedings of the ACM on Measurement and Analysis of Computing Systems (Sigmetrics) 2021, no. 1: 1-29. (Acceptance Rate: 12%)

[16] **Lishan Yang**, Bin Nie, Adwait Jog, Evgenia Smirni, "Practical Resilience Analysis of GPGPU Applications in the Presence of Single- and Multi-bit Faults", IEEE Transactions on Computers, vol. 70, no. 1, pp. 30-44, 1 Jan. 2021.

[17] Bin Nie, **Lishan Yang**, Adwait Jog, Evgenia Smirni, "Fault Site Pruning for Practical Reliability Analysis of GPGPU Applications", 51st Annual IEEE/ACM International Symposium on Microarchitecture (MICRO) 2018: pp. 749-761. (Acceptance Rate: 21%)

[18] Kai Zhang, Bingsheng He, Jiayu Hu, Zeke Wang, Bei Hua, Jiayi Meng, **Lishan Yang**, "G-NET: Effective GPU Sharing in NFV Systems", 15th USENIX Symposium on Networked Systems Design and Implementation (NSDI) 2018: pp. 187-200. (Acceptance Rate: 16%)

[19] **Lishan Yang,** Ludmila Cherkasova, Rajeev Badgujar, Jack Blancaflor, Rahul Konde, Jason Mills, Evgenia Smirni, "Evaluating Scalability and Performance of a Security Management Solution in Large Virtualized Environments", Proceedings of the 2018 ACM/SPEC International Conference on Performance Engineering (ICPE): pp. 168-175.

Refereed Posters

[1] **Lishan Yang**, George Papadimitriou, Dimitrios Sartzetakis, Adwait Jog, Evgenia Smirni, Dimitris Gizopoulos, "Probing Weaknesses in GPU Reliability Assessment: A Cross-Layer Approach", IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS) 2024, Poster track.

[2] **Lishan Yang**, "Typhoon: Enabling GPGPU Application Resilience Estimation with Different Input Types", ACM Sigmetrics Student Research Competition (SRC) 2021

[3] **Lishan Yang**, Anna Schmedding, Riccardo Pinciroli, Evgenia Smirni, "Simulating COVID-19 Containment Measures Using the South Korean Patient Data", Sensys 2020, COVID-19 Pandemic Research Poster Session.

STUDENTS

Ph.D. Students

Yu Sun	George Mason University, since 2023 Fall
Daehyun Lee	George Mason University, since 2026 Fall M.S., 2024-2026, George Mason University
Ying Tang	George Mason University, since 2025 Fall
Giuseppe De Rosa	Visiting Ph.D. student from Scuola IMT Alti Studi Lucca since 2025 Spring

M.S. Students

Rohan Mengle	George Mason University, since 2026 Spring Undergrad student, 2025 Fall, George Mason University
Ming Tang	George Mason University, since 2026 Spring
Zhu Zhu	George Mason University, 2023-2026

Undergraduate Students

Sam O'Neill	George Mason University, 2025
Dhatri Parakal	University of Illinois Urbana-Champaign (UIUC), 2024-2025 recruited from Aspiring Scientists Summer Internship Program (ASSIP)
Cherish Mulpuru	George Mason University, 2024-2025 supported by CCI NoVA-node URA program
Jagan Yetukuri	George Mason University, 2024-2025
Weilon Price	George Mason University, 2022-2023

High School Students

(Through an initiative of "Directed Reading" program with Holton-Arms School)

Zara Ezebuiro	2026-Present Reading Topic: Operating Systems
Tian Feng	2026-Present Reading Topic: Operating Systems
Sofia Francisco Skamnelos	2024-2025 Reading Topic: Operating Systems
Grace Xiao	2024-2025 Reading Topic: Operating Systems
Lucia M. Noto	2024-2025 Reading Topic: Monte Carlo Simulation

SERVICE

Program Committee Chairs

2027	PC Associate Chair	The 57th Annual IEEE/IFIP International Conference on Dependable Systems and Networks (DSN)
2027	Workshop Chair	International Workshop on Teaching Performance Analysis of Computer Systems (TeaPACS)
2023	PC Chair	Dependable and Secure Machine Learning Workshop (DSML)

Program Committees

2027 ASPLOS, DSN, FORGE

2026 Sigmetrics, DSN, PPoPP, HPDC, IPDPS, IISWC, EDCC
HPDC Poster, Sigmetrics SRC, ACM SRC Grand Final, DSN Doctoral Symposium, QA4Agent workshop

2025 Sigmetrics, DSN, PPoPP, ICPE, SRDS, ICDCS
DT4DRS workshop, U-SPACE workshop, FSE SRC, SC SRC

2024 Sigmetrics, DSN, ICPE, CLUSTER, SAC
CCGrid Doctoral Symposium and Early Career

2023 DSN, SC, IPDPS, HPDC, SRDS
AIPerf workshop, LADC Student Forum, ESANN Special Session

Organizing Committees

2027 DSN Publication Chair, ICPE Artifact Evaluation Track Chair

2026 Sigmetrics Student Activities Chair

2025 Sigmetrics SRC Chair, DSN Artifact Chair, ICPE Industry Track Chair

2024 Sigmetrics SRC Chair, DSN Posters Track Chair, HPDC Social Media Chair, ICPE Vision and WiP Track Chair

2023 Sigmetrics Publicity Chair, HPDC Social Media Chair, HotStorage Virtual Conference Chair, DSML General Chair

Steering Committee

2023-Present Dependable and Secure Machine Learning Workshop (DSML)

Journal Reviewing

- IEEE Transactions on Computers (TC)
- ACM Transactions on Modeling and Performance Evaluation of Computing Systems (TOMPECS)
- IEEE Transactions on Dependable and Secure Computing (TDSC)
- IEEE Transactions on Cloud Computing (TCC)
- IEEE Transactions on Reliability
- IEEE/ACM Transactions on Networking (ToC)
- IEEE Transactions on Parallel and Distributed Systems (TPDS)

Panelists

2026 The Sixth Workshop on Benchmarking and Performance of Machine Learning Workloads on Emerging Hardware (MLBENCH)
2024 ISSRE Doctoral Symposium Discussion
2022 W&M Symposium for Graduate Studies

Outreach Activities

2025-Present Organizing Committee: Virginia Women in High-Performance Computing (VA-WHPC)
2023-Present Faculty Advisor: GMU CS-GSA (Graduate Student Association)
2022-2025 SC Mentor-Protégé Program
2026 DSN Student Mentoring Program
2026 Sigmetrics Student Mentoring Program

TEACHING

- Modern Computer Architecture
  CS665 (2026 Fall)
- Computer Systems Performance Evaluation
  CS672 (2026 Spring)
- Special Topics: Advanced Computer Architecture
  CS692 (2025 Spring, 2024 Spring), CS 695 (2023 Spring)
- Special Topics: Reliability in Computer Systems
  CS 695 (2024 Fall)
- Concurrent & Distributed Systems
  CS 475 (2023 Fall, 2022 Fall)